SYLLABUS
STAT 511: MATHEMATICAL STATISTICS
FALL 2024

UNIVERSITY OF ILLINOIS AT URBANA CHAMPAIGN

Course Number: STAT 511
Title: Mathematical Statistics
Credit Hours: 4.0 graduate credit hours
Instructor: Sabyasachi Chatterjee. Email: sc1706@illinois.edu
Teaching Assistant: Kaustav Chakraborty.
Lecture Hours: Tuesdays and Thursdays 9:30-10:50 AM.

Office Hours:

Course description and Goals: This course is a second course on Mathematical Statistics and Statistical Inference and will cover the fundamentals of theoretical statistics. Topics include: concentration of measure, basic empirical process theory, convergence, point and interval estimation, maximum likelihood, hypothesis testing, Bayesian inference, nonparametric statistics and bootstrap resampling. This course is excellent preparation for advanced work in Statistics and Machine Learning.

Prerequisites: I assume that you are familiar with basic probability and mathematical statistics. You should already know the following concepts: probability, distribution functions, density functions, moments, transformation of variables, and moments generating functions. This course moves extremely fast. Depending on your background and maturity level, it might take significant effort from your side to keep up with the class. Being up to date and asking questions is the best way to learn from this course.

Lists of Readings and Texts: The main reference for this class is the textbook "All of Statistics" by Larry Wasserman. I will include a soft copy on the course website. We will tentatively cover Chapters $1-12$ and some additional material. The following course website https://www.stat.cmu.edu/~larry/=stat705/ in Larry Wasserman's home page contains lecture notes and homeworks. You can take a look. I will follow these closely.

Homework: There will be weekly homeworks. Homework assignments will be posted, roughly weekly, on Canvas. You should submit your HW on Canvas by scanning and uploading it. If you need an extension due to illness, email me BEFORE the homework deadline. The homeworks are meant for you to practice solving problems. Do not search for HW solutions online. You may work with others on the homework. But write-up your final solutions on your own. Please credit the students that you work with clearly.

Date: 26 August, 2024.

Exams: There will be a mid and a end term exam.

Grading Scale:

(1) Homeworks: 40 percent.
(2) Exam 1: 25 percent.
(3) Exam 2: 25 percent.
(4) Scribing: 10 percent.